

Designing an experiment

MATH 213

Group 1

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart’s store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

- 1. What are the cases for your experiment? **The sample of students who participate in the experiment.**
- 2. What variables are you recording about those cases? **Which soda students prefer, how much do you drink soda**
- 3. To make sure you’re thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Name of students	preference
James	Dr. Pepper
Alliyah	Dr. Thunder

4. How will select your sample of students that reduces (or eliminates) sampling bias?

Having a random sign up for people to join in on the experiment.

5. What are some possible confounding variables that could affect this taste test?

Age, food preferences, health, hunger, level of sense of smell, state of the drink, drank soda already? What soda is normally preferred?

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?
If they drink one of the types of soda everyday. If they don't even drink soda. The can, also the knowledge of which drink one is.

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

Have people sign up to test the difference between the sodas. The participants are given two cups, one with Dr. Pepper and one with Dr. Thunder, and not told which is in which cup. Only the experimenter will know via a number on the cup. The participant will be asked about their age, food preferences, health, hunger, level of sense of smell, state of the drink, if they drank soda already, and if so what soda preferences they have. After the participant samples both cups, they will be asked which cup they prefer, either 1, 2, or if they liked both.

Group 2

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

- 1. What are the cases for your experiment?
N number of students
- 2. What variables are you recording about those cases?
Preference of soda A/B
Group 1/2 to determine which soda they got first
- 3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Cases	Preference of Soda (A/B)	Group (1/2) (Which drink first)
Case 1	A	1
Case 2	B	2

4. How will select your sample of students that reduces (or eliminates) sampling bias?

Access database of students, randomly picking student ID numbers, and inviting them for the taste test

5. What are some possible confounding variables that could affect this taste test?

Something affecting taste such as prior meals/soda drinking/gum/chapstick/etc.

Sodas being in different condition (flatter/warmer than the other)

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

If they already drink a lot of either soda and would already have a preference

People may prefer the first / last soda just by nature of it being given first / last (randomize order soda is given)

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.
 1. **Hack into hood college student database to find list of all students and randomly select n amount of students by ID number. Evenly sort them into two groups to determine which drink they get first**
 2. **Send guinea pigs email inviting them to our random drink taste test, asking them to fast for an hour before**
 3. **Set up a stand indoors where the subject can sit down comfortably**
 4. **Present them a palette cleanser of room temperature filtered water because no way anyone read instructions to fast beforehand + to make sure they aren't thirsty**
 5. **Go behind a curtain and pour a fresh glass of Drink A, make student drink**
 6. **Repeat palette cleanser**
 7. **Go behind curtain and pour fresh glass of Drink B, make student drink**
 8. **Repeat palette cleanser**
 9. **Ask them for preference between both drinks.**
 10. **Send them home with a pat on the back and a lollipop**

Group 3

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

- 1. What are the cases for your experiment?
of hood college students in the experiment
- 2. What variables are you recording about those cases?

Student ID, Year and preference (Dr.Pepper or Dr.Thunder)

- 3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Student ID	Year in school	preference
1001	junior	Dr. Pepper
1002	senior	Dr. Thunder

4. How will select your sample of students that reduces (or eliminates) sampling bias?

Placing the student ID numbers into a random number generator and having it select, n amount of students

5. What are some possible confounding variables that could affect this taste test?

What they had to eat prior, dislike of soda in general and how thirsty the student is

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

Hiding the labels on the cans

How thirsty the student is

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

Select a number of students using the random number generator, to ensure that it is a simple random sample.

Remove/cover the labels on the cans to prevent bias on the students end

Ask the student what they had to eat prior, if they dislike of soda and how thirsty the student is (1-5) and record.

Have them sample the first soda (Dr.Pepper), and have them drink water after to ensure a clean palette

Repeat

Ask which soda they preferred (1 or 2).

Group 4

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart’s store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

- 1. **What are the cases for your experiment?** Students
- 2. **What variables are you recording about those cases?** Choice/Preference, How often they drink soda(1 (never) - 5 (daily)), Favorite soda(?)
- 3. **To make sure you’re thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.**

	Choice/Preference	How often do you drink soda	Favorite soda	First Drink
Student 1	Name-brand	2	Crush	Name-brand
Student 2	Name-brand	5	A&W	Store-brand
Student 3	Store-brand	3	Aldi cola	Store-brand

4. How will you select your sample of students that reduces (or eliminates) sampling bias?

Conduct experiment in each building on campus with simple random sampling (AKA stratified sampling)

5. What are some possible confounding variables that could affect this taste test?

Order that they drink each soda in

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

Preference for choosing people you know, amount of soda given

7. **Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.**
 1. Go to each building(+ commuter lounge) and randomly select 10 students each
 - a. Place every possible room number in a hat and randomly select 10
 - b. Choose a random student from that room
 2. Randomly give the name-brand or store-brand first
 - a. Double blind
 3. Give water to clear palette
 4. Give second soda
 5. Ask for preference, then favorite soda and how often they drink soda

Group 5

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart’s store-brand Dr. Thunder.
Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

1. What are the cases for your experiment?

The random students at Hood College selected for the sample

2. What variables are you recording about those cases?

Whether the students preferred Dr. Pepper or Dr. Thunder (categorical), whether they are healthy or sick (categorical), whether they recently ate (categorical), whether they like soda or not (categorical)

3. To make sure you’re thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Student #	Preference	Health	RecentlyAte	SodaOpinion
1	Dr. Pepper	Healthy	No	Likes Soda
2	Dr. Thunder	Healthy	Yes	Likes Soda

4. How will select your sample of students that reduces (or eliminates) sampling bias?

We select random students by placing names in a spreadsheet and have a random number generator select the names of the students we include in the sample.

5. What are some possible confounding variables that could affect this taste test?

Possible confounding variables could be:

- The student already had their mind made up or they haven't given one brand a fair shot.
- If the student already ate before hand.
- If the student is sick when taking the taste test.
- If the student has a dislike of soda in general.

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

Hiding the brand from the students would reduce bias. Asking the students if they are sick, asking when they last ate, and asking if they like soda would also ensure their preference answer is accurate.

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

When selecting our sample from students at Hood College, we first put all the names of Hood students into an excel spreadsheets and have a random number generator select 100 random names.

We then track down those 100 students and provide them with 2 cans of soda with the brands hidden. Before they try the sodas we ask them some questions and record it in the spreadsheet. These questions are: Are you currently sick? Have you ate anything recently? Do you like soda in general?

After recording their answers, we have them try the sodas and record which soda they prefer.

Group 6

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

1. What are the cases for your experiment?
2. What variables are you recording about those cases?
3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

4. How will select your sample of students that reduces (or eliminates) sampling bias?
5. What are some possible confounding variables that could affect this taste test?
6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

Group 7

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

1. What are the cases for your experiment?

Students that use the beverage vending machine weekly.

2. What variables are you recording about those cases?
 - Student name
 - Which soda they like more,
 - Do they drink soda?
 - Do they taste the same (1- Strong No, 5- Strong Yes)
3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Name	Do they drink soda	Which do they like more	Do they taste different (1 - 5)
Jimmy	No	Dr. Thunder	2
Timmy	Yes	Dr. Pepper	4

4. How will select your sample of students that reduces (or eliminates) sampling bias?

It will sample students that use a vending machine that has both Dr. Pepper and Dr. Thunder products.

5. What are some possible confounding variables that could affect this taste test?

- They do drink soda but they have never tasted Dr. Pepper or Dr. Thunder
- The vending machine workers let the drinks expire.
-

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

- Being allergic to the ingredients in one vs the other.
- Vendetta against off brand soda

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

We wish to determine which drink students that use the vending machine weekly prefer, whether it is Dr. Thunder or Dr. Pepper. We have these students answer questions and tally results on their preferred drink. They answer questions on a bulletin board placed next to the vending machine. The questions are asking for your name, if you drink soda, your preferred drink between dr. Thunder and Dr. Pepper, as well as if you think they taste similar on a scale from 1 to 5.-+

Group 8

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

1. What are the cases for your experiment? The Hood students that tried the soda
2. What variables are you recording about those cases?

Does each case like Dr. Pepper or Dr. Thunder more?

3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Students	Preference
Student 1	Dr. pepper
Student 2	Dr. Pepper
Student 3	Dr. Thunder

4. How will select your sample of students that reduces (or eliminates) sampling bias?
 - Look at alphabetical list of all enrolled students (ie: like an email list)
 - Assign a number to each name
 - Randomly generate (pseudo-randomly or truly randomly) 'n' numbers
 - Select the names only with the numbers selected
 - Reach out to these students asking if they would be interested in blind tasting sodas (can email based off of the email list) and telling which is better, perhaps offering a free can of whichever to increase response rates.

5. What are some possible confounding variables that could affect this taste test?

Some confounding variables that could possibly affect this experiment could be, if they have eaten or drank anything recently, to fix this confounding variable you might want to provide water so that the subjects can cleanse their palate before testing the sodas

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

A possible source of bias that could arise in the taste test could be any affiliations that the subject has with the company, for example if one of the students has a side job at the Dr. Pepper manufacturing plant.

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.
1. Buy several cases of both classic Dr. Pepper and classic Dr. Thunder (at least 60 cans of each)
2. Buy styrofoam cups to enable blinding (subject will not know which soda is in which cup)
3. Using the Hood college email database, create a spreadsheet of every name in said database, assign numbers to each name, and randomly select a number.
4. Email the selected student, asking if they would be interested in the study, ensuring their identity will remain anonymous and offering a free unopened can of whichever soda they decide they like more.
5. Repeat steps 3-4 until a sample size of 30 is achieved.
6. Conduct the experiment, pour three cups (single blind, the researcher must know the soda for recording purposes, but the subject should not, can be achieved by arbitrarily marking the cups of a certain type of soda so the researcher knows), two containing the sodas, one containing water. Ask them to take a drink of one soda, followed by a drink of water, followed by a drink of the second soda.
7. Ask them which soda they prefer. They can then choose to drink as much more soda and/or water as they like before coming to a conclusion.
8. Once the subject comes to a conclusion about which soda they liked more, give them a can of whatever that soda may be, record their data into a spreadsheet, and interpret results.

Group 9

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

1. What are the cases for your experiment?
50 Hood students
2. What variables are you recording about those cases?
 - Favorite soda
 - Average amount of cans of soda drank per week (convert bottles or other to equivalent amount of cans)
 - Order of soda (A is Dr. Pepper first, B is Dr. Thunder first)
 - Do you prefer Dr. Pepper or Dr. Thunder?
 - How many times a month do you go to Walmart?
3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Case	Order of tasting	Favorite soda	Average Cans	# Times at Walmart/Month	Pepper or Thunder
Student 1	A	Pepsi	7	4	Pepper
Student 2	B	Fanta	2	0	Thunder

4. How will select your sample of students that reduces (or eliminates) sampling bias?

Random sample of 25 Hood residents and 25 Hood commuters

Also, cover labels on the cans during taste test, or pour both into different (solid, not see-through) cup

5. What are some possible confounding variables that could affect this taste test?

Students may have a meal plan, Hood has Dr. Pepper

Students who do not have vehicles may not have transportation to Walmart

Order of trying Pepper or Thunder first

6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?

Availability, cost (meal plan), cost (between sodas), researchers recording results do not know which group participants are in, ~~exclude people who say favorite soda is Dr. Pepper or Dr. Thunder because they may not honestly choose between the two~~ those who love Fanta, for example, may be more reliable in determining which tastes better. (no, because not reflective of student preferences)

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.
 - Use database of Hood students to filter residents and commuters, and randomly select from them
 - Survey questions
 - Taste test, choose Dr. Pepper or Dr. Thunder
 -

Group 10

Imagine that you want to conduct an experiment to determine whether students at Hood prefer name-brand Dr. Pepper or Walmart's store-brand Dr. Thunder.

Instead of doing a census – that is, to collecting data from every student at Hood – assume that you will need to select a sample from the population.

- 1. What are the cases for your experiment?
All individuals (student and staff) that submit to the experiment and perform the taste test.
- 2. What variables are you recording about those cases?
Which drink do they prefer in each randomized taste test?
- 3. To make sure you're thinking about this thoroughly, mock up at least two rows of what your dataframe will look like (including some made-up data) using the Insert menu above and selecting Table.

Case	Box 1 (A/B)	Box 2 (A/B)	Box 3 (A/B)	Favorite Soda	Age Group
Jon	A	B	A	Cream Soda	30-40
Jose	B	B	A	Coke	18-25

4. How will select your sample of students that reduces (or eliminates) sampling bias?
Ask students for their favorite soda and eliminate students that have a stated preference for the tested sodas.
5. What are some possible confounding variables that could affect this taste test?
Are they eating anything at time of sample?
Are they properly hydrated?
Are there any factors affecting their sense of taste?
Age of case as they are more likely to be able to afford the non-knock off brand and have preferences as a result.
6. What are some other possible sources of bias in this taste test that you may want to try to eliminate?
Sampling bias because we're only testing individuals passing through Hood's Whitaker Center

7. Describe your experiment from start to finish, making sure to include all the details about avoiding pitfalls from the previous slides.

Set up in Whitaker Center with the three boxes of blind taste testing. Randomize the order of the boxes so the testers don't know which boxes have A/B of which sample (double blind).

Any individuals that pass through Whitaker Center are asked to participate in the taste test with their answers being recorded.